

Audio Physical Dynamics Inspired Deepfake Detection for Voice Authentication Systems

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Abstract

Voice biometric authentication systems are increasingly targeted by voice cloning spoof attacks. This letter presents a physical-dynamics inspired feature extraction method that measures vocal fold subglottal pressure variation and vocal tract transfer function boundaries to differentiate organic human speech from AI-synthesized signals.

Key Methodology & Architecture

Glottal flow waveform estimation, Linear Predictive Coding (LPC) residual analysis, Bi-LSTM classifier, biometric score fusion.

Datasets & Benchmark Environments

Logical Access Spoofing Dataset, Telephony Voice Banking Test Set.

Performance Metrics & Graph Axis Parameters

Reported Results: Equal Error Rate (EER) = 0.65%, Verification Latency = 8.5ms, Spoof Rejection Rate = 99.35%.

Graph Parameter Mapping: X-axis: Biometric Threshold Score (-1.0 to 1.0); Y-axis: False Acceptance Rate (FAR) & False Rejection Rate (FRR).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Low latency suitable for real-time mobile banking and voice assistant security.	Requires high audio sampling rate (min 16kHz) for glottal residual extraction.

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